

Leaf Stomata Microscope Detective

Compare standardized stomatal density without overclaiming water use

At a glance: Compare stomatal density with a standardized count and make a cautious, testable hypothesis. Plan about 80 minutes for the core block; 4 teams of 4 students, one digital microscope per team.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: stomata, water loss, gas exchange, microscopy, data sampling (Understand).
- Apply one standardized surface, preparation, magnification, field area, and counting rule across samples (Apply).
- Calculate and compare mean stomatal density and range from at least three fields per sample (Analyze).
- Defend a cautious hypothesis while stating that counts alone cannot rank actual water use (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- leaf samples: 3 to 4 fresh leaves per team from contrasting habitats (about 24 total, collect on campus the morning of)
- clear tape: clear packing/transparent tape, a few short strips per team (2 rolls for the camp)
- clear nail polish: 1 small bottle covers all 4 teams (bring 2; ventilated station only)
- glass slides: about 16 per run; a 50 to 72 slide box for the camp
- prepared stomata slides: about 4 as a backup for failed peels (order a small set or borrow)
- digital microscopes: one per team (4; two on hand, rotate or borrow two more); screen models need charging, USB-cable models need a laptop/tablet
- grid/counting sheets: 1 per team or per student (20 with buffer); counting sheet in this guide
- gloves: latex-free, a box at the polish station (optional, not class-wide)

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Set up one station per team with the materials above, sorted and ready.
- 02 Stage the scoring and counting sheets plus 4 backup prepared slides. Fix the same leaf surface, preparation, magnification, field area, and fully-inside counting rule for all samples; require at least three fields and both mean and range.
- 03 Set up one ventilated polish-and-dry station with the nail polish, a box of latex-free gloves, and a drying tray, away from the microscope tables. No goggles or containment trays are needed: this is a dry counting activity with one small ventilated polish step.
- 04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Compare stomatal density with one standardized method, then make a cautious hypothesis." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before preparing slides, each team predicts which sample will have the highest stomatal density and gives one reason.

Phase 3 Constraints

0:12 - 0:18

Use the same leaf surface, preparation, magnification, field area, and counting rule for every sample. Count at least three fields and report mean plus range; there is no redesign round.

Phase 4 Make impressions and count

0:18 - 0:44

Teams prepare impressions at the ventilated station, then count at least three equal fields per sample using the fixed surface, preparation, magnification, field area, and fully-inside rule. Circulate to verify the standard without counting for them.

Phase 5 Calculate and compare (within the work block)

Teams calculate mean and range for each sample and compare stomatal density. They may state a cautious hypothesis, but counts alone cannot rank actual water use.

Phase 6 Add evidence or test a surface (extension)

Teams who finish add a fourth standardized field or test the other surface as a separately labeled comparison, then recompute mean and range without claiming actual water use.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, defend the density comparison with mean and range, state the counts-only limitation, then reset the station, cap polish, wipe tables, and wash hands after removing gloves if worn.

TROUBLESHOOTING

Problem: Counts vary widely between fields. **Fix:** Recheck the fixed surface, magnification, field area, and fully-inside rule. Keep all valid fields and report the range with the mean rather than hiding variation.

Problem: The peel shows nothing under the microscope. **Fix:** Confirm the polish dried fully and the intended surface was sampled. If it failed, use a backup prepared slide so the team can still compare standardized density.

Problem: Stomata are hard to find or out of focus. **Fix:** Start at low magnification to locate the impression, then increase magnification and re-focus; make sure the taped peel is flat with no bubbles, and re-make the peel if it is smeared.

Problem: A team finishes early. **Fix:** Add a fourth standardized field, recompute mean and range, and explain whether the extra field changes the cautious hypothesis.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to stomata: pores for gas exchange.

Tier 2 (counting but not reasoning):

- "Which parts of the sampling method must stay identical for a density comparison?"

Tier 3 (ready to extend):

- "What aperture, pore-size, gas-exchange, species, and environmental data would be needed before discussing actual water use?"

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Why count several fields instead of one?
- Why can stomatal counts alone not determine a leaf's actual water use?
- Where might error sneak into your count?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Slide or peel quality	25
Standardized counting accuracy	30
Cautious evidence-backed hypothesis	25
Mean-and-range data table	10
Cleanup and care	10
Total	100

CLEANUP AND SOURCES

Return microscopes and slides, dispose of used tape and peels, cap polish at the ventilated station, wipe tables, wash hands, and collect score slips.

SOURCES California Academy of Sciences, Stomata Printing microscope investigation; Rothamsted Bioimaging, Measuring Stomatal Density using nail varnish; Bertolino, Caine, and Gray (2019), PMID PMC6414756; Lunn et al. (2024), PMID PMC11565199; USDA Forest Service hardwood physiology research.